SQL JOINS: 40 Practice Questions

Multiple choice (4 options each) with practice dataset and answer key

Instructions

1. Run the SQL script in Section 2 to create the practice database (MySQL).
2. Covers INNER JOIN, LEFT JOIN, RIGHT JOIN and CROSS JOIN only. Questions 1–15 test concepts. Questions 16–40 use the dataset; run the query yourself if you are unsure.
3. Each question has exactly one correct answer. The answer key with explanations is at the end.

Section 1: Dataset

The database has 5 related tables. NULL values are shown in red.

Table: departments

dept_id	dept_name	location
1	HR	Delhi
2	Sales	Mumbai
3	IT	Bangalore
4	Finance	Delhi
5	Marketing	Pune

Marketing (dept_id 5) has no employees.

Table: employees

emp_id	emp_name	dept_id	salary
101	Rahul	3	90000
102	Priya	3	70000
103	Ankit	2	55000
104	Sneha	2	48000
105	Vikas	1	45000
106	Neha	4	65000
107	Arjun	*NULL*	40000
108	Kavya	3	72000

Arjun has no department (dept_id is NULL). Only Ankit and Sneha act as salespersons in orders.

Table: customers

customer_id	customer_name	city
1	Ravi	Delhi
2	Meena	Mumbai
3	Suresh	Pune
4	Anjali	Delhi
5	Karan	Jaipur
6	Pooja	Mumbai

Karan and Pooja have never placed an order.

Table: products

product_id	product_name	category	price
201	Laptop	Electronics	50000
202	Mobile	Electronics	20000
203	Chair	Furniture	3000
204	Desk	Furniture	7000
205	Headphones	Electronics	2000

Headphones has never been ordered.

Table: orders

order_id	customer_id	product_id	emp_id	quantity	order_date
1001	1	201	103	1	2024-01-05
1002	1	202	104	2	2024-01-10
1003	2	203	103	4	2024-02-02
1004	3	201	104	1	2024-02-15
1005	4	204	103	2	2024-03-01
1006	2	202	104	1	2024-03-12
1007	7	203	*NULL*	3	2024-03-20
1008	1	204	103	1	2024-04-02

Order 1007 belongs to customer_id 7, which does not exist in customers, and has no salesperson (emp_id is NULL).

How the tables connect

- `employees.dept_id → departments.dept_id`
- `orders.customer_id → customers.customer_id`
- `orders.product_id → products.product_id`
- `orders.emp_id → employees.emp_id (salesperson)`

Section 2: SQL Script (MySQL)

Copy and run this script to create and fill the tables.

```
CREATE DATABASE IF NOT EXISTS joins_practice;
USE joins_practice;

CREATE TABLE departments (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(30),
    location VARCHAR(30)
);
INSERT INTO departments VALUES
(1, 'HR', 'Delhi'),
(2, 'Sales', 'Mumbai'),
(3, 'IT', 'Bangalore'),
(4, 'Finance', 'Delhi'),
(5, 'Marketing', 'Pune');

CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(30),
    dept_id INT,
    salary INT
);
INSERT INTO employees VALUES
(101, 'Rahul', 3, 90000),
(102, 'Priya', 3, 70000),
(103, 'Ankit', 2, 55000),
(104, 'Sneha', 2, 48000),
(105, 'Vikas', 1, 45000),
(106, 'Neha', 4, 65000),
(107, 'Arjun', NULL, 40000),
(108, 'Kavya', 3, 72000);

CREATE TABLE customers (
    customer_id INT PRIMARY KEY,
    customer_name VARCHAR(30),
    city VARCHAR(30)
);
INSERT INTO customers VALUES
(1, 'Ravi', 'Delhi'),
(2, 'Meena', 'Mumbai'),
(3, 'Suresh', 'Pune'),
(4, 'Anjali', 'Delhi'),
(5, 'Karan', 'Jaipur'),
(6, 'Pooja', 'Mumbai');

CREATE TABLE products (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(30),
    category VARCHAR(30),
    price INT
);
INSERT INTO products VALUES
(201, 'Laptop', 'Electronics', 50000),
(202, 'Mobile', 'Electronics', 20000),
(203, 'Chair', 'Furniture', 3000),
(204, 'Desk', 'Furniture', 7000),
(205, 'Headphones', 'Electronics', 2000);

CREATE TABLE orders (
    order_id INT PRIMARY KEY,
    customer_id INT,
```

```
    product_id INT,
    emp_id INT,
    quantity INT,
    order_date DATE
);
INSERT INTO orders VALUES
(1001, 1, 201, 103, 1, '2024-01-05'),
(1002, 1, 202, 104, 2, '2024-01-10'),
(1003, 2, 203, 103, 4, '2024-02-02'),
(1004, 3, 201, 104, 1, '2024-02-15'),
(1005, 4, 204, 103, 2, '2024-03-01'),
(1006, 2, 202, 104, 1, '2024-03-12'),
(1007, 7, 203, NULL, 3, '2024-03-20'),
(1008, 1, 204, 103, 1, '2024-04-02');
```

Section 3: Questions

Part A: Concepts (Q1–Q15)

Q1. Which JOIN returns only the rows that have matching values in both tables?

A) LEFT JOIN
B) INNER JOIN
C) RIGHT JOIN
D) CROSS JOIN

Q2. What does a LEFT JOIN return?

A) All rows from the left table, plus matching rows from the right table (NULL where there is no match)
B) All rows from the right table and matching rows from the left table
C) Only matching rows from both tables
D) Every combination of rows from both tables

Q3. Which JOIN returns all rows from the right table, plus matching rows from the left table (NULL where there is no match)?

A) CROSS JOIN
B) INNER JOIN
C) LEFT JOIN
D) RIGHT JOIN

Q4. Table A has 6 rows and table B has 4 rows. How many rows does A CROSS JOIN B return?

A) 10
B) 6
C) 4
D) 24

Q5. Which JOIN is written WITHOUT an ON condition?

A) INNER JOIN
B) LEFT JOIN
C) CROSS JOIN
D) RIGHT JOIN

Q6. Which keyword is used to write the join condition?

A) ON
B) HAVING
C) WHERE
D) GROUP BY

Q7. If you write only JOIN (without INNER/LEFT/RIGHT), which join is performed?

A) LEFT JOIN
B) CROSS JOIN
C) INNER JOIN
D) RIGHT JOIN

Q8. Table A has 5 rows and table B has 3 rows. Every row of A matches exactly one row of B. How many rows does A INNER JOIN B return?

A) 5
B) 3
C) 8
D) 15

Q9. Which query pattern finds rows in table A that have NO matching row in table B?

A) A INNER JOIN B ON A.id = B.id
B) A LEFT JOIN B ON A.id = B.id WHERE A.id IS NULL
C) A CROSS JOIN B WHERE B.id IS NULL
D) A LEFT JOIN B ON A.id = B.id WHERE B.id IS NULL

Q10. A RIGHT JOIN B gives the same rows as:

A) B CROSS JOIN A
B) A LEFT JOIN B
C) A INNER JOIN B
D) B LEFT JOIN A

Q11. Table A has 10 rows. What is the MINIMUM number of rows that A LEFT JOIN B can return?

A) 0
B) 10
C) Same as the number of rows in B
D) 1

Q12. In an INNER JOIN, what happens to a row whose join column value is NULL?

A) It matches every row of the other table
B) It matches other rows that also have NULL
C) It is never matched, so it is excluded from the result
D) The query gives an error

Q13. Why do we use table aliases (e.g. FROM employees e) in join queries?

A) To make the query run faster
B) Because joins cannot work without aliases
C) To shorten table names and make clear which table each column comes from
D) To rename the table permanently in the database

Q14. FROM a CROSS JOIN b WHERE a.id = b.id gives the same result as:

A) a INNER JOIN b ON a.id = b.id
B) a RIGHT JOIN b ON a.id = b.id
C) a LEFT JOIN b ON a.id = b.id
D) a CROSS JOIN b (no change)

Q15. In a LEFT JOIN, putting a filter on a right-table column in the WHERE clause (e.g. WHERE o.quantity > 1) usually:

A) Has no effect on the result
B) Removes the unmatched left rows, so it behaves like an INNER JOIN
C) Turns the query into a CROSS JOIN
D) Causes a syntax error

Part B: Dataset-Based (Q16–Q40)

Q16. How many rows does this query return?

```
SELECT e.emp_name, d.dept_name
FROM employees e
INNER JOIN departments d ON e.dept_id = d.dept_id;
```

A) 5
B) 8
C) 7
D) 9

Q17. **How many rows does this query return?**

```
SELECT e.emp_name, d.dept_name
FROM employees e
LEFT JOIN departments d ON e.dept_id = d.dept_id;
```

A) 7
B) 5
C) 8
D) 13

Q18. **In the result of the query in Q17, which employee has NULL as dept_name?**

A) Rahul
B) Kavya
C) Arjun
D) Vikas

Q19. **How many rows does this query return?**

```
SELECT d.dept_name, e.emp_name
FROM departments d
LEFT JOIN employees e ON d.dept_id = e.dept_id;
```

A) 5
B) 8
C) 7
D) 9

Q20. **Which department is returned by this query?**

```
SELECT d.dept_name
FROM departments d
LEFT JOIN employees e ON d.dept_id = e.dept_id
WHERE e.emp_id IS NULL;
```

A) Marketing
B) Finance
C) HR
D) No department is returned

Q21. **How many rows does this query return?**

```
SELECT c.customer_name, o.order_id
FROM orders o
INNER JOIN customers c ON o.customer_id = c.customer_id;
```

A) 8
B) 7
C) 6
D) 10

Q22. **How many customers have never placed an order?**

```
SELECT c.customer_name
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
WHERE o.order_id IS NULL;
```

A) 0
B) 1
C) 3
D) 2

Q23. **How many rows does this query return?**

```
SELECT c.customer_name, o.order_id
```

```
FROM customers c
RIGHT JOIN orders o ON c.customer_id = o.customer_id;
```

A) 6
B) 8
C) 7
D) 10

Q24. How many rows does this query return?

```
SELECT p.product_name, o.order_id
FROM orders o
RIGHT JOIN products p ON o.product_id = p.product_id;
```

A) 8
B) 13
C) 5
D) 9

Q25. How many rows does this query return?

```
SELECT d.dept_name, p.product_name
FROM departments d
CROSS JOIN products p;
```

A) 10
B) 25
C) 5
D) 0

Q26. How many rows does this query return?

```
SELECT c.customer_name, p.product_name
FROM customers c
CROSS JOIN products p
WHERE c.city = 'Delhi';
```

A) 10
B) 5
C) 2
D) 30

Q27. How many employees have NOT handled any order?

```
SELECT e.emp_name
FROM employees e
LEFT JOIN orders o ON e.emp_id = o.emp_id
WHERE o.order_id IS NULL;
```

A) 2
B) 0
C) 8
D) 6

Q28. Which order is returned by this query?

```
SELECT o.order_id
FROM employees e
RIGHT JOIN orders o ON e.emp_id = o.emp_id
WHERE e.emp_id IS NULL;
```

A) 1001
B) 1004
C) No order
D) 1007

Q29. What is the total quantity sold in the Furniture category?

```
SELECT p.category, SUM(o.quantity) AS total_qty
FROM products p
INNER JOIN orders o ON p.product_id = o.product_id
GROUP BY p.category;
```

A) 7
B) 3
C) 10
D) 13

Q30. What is the total quantity ordered by the customer Ravi?

```
SELECT SUM(o.quantity)
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
WHERE c.customer_name = 'Ravi';
```

A) 3
B) 5
C) 2
D) 4

Q31. Which product has never been ordered?

```
SELECT p.product_name
FROM products p
LEFT JOIN orders o ON p.product_id = o.product_id
WHERE o.order_id IS NULL;
```

A) Desk
B) Headphones
C) Chair
D) Mobile

Q32. What is the total revenue (quantity × price) from Electronics products?

```
SELECT SUM(o.quantity * p.price)
FROM orders o
JOIN products p ON o.product_id = p.product_id
WHERE p.category = 'Electronics';
```

A) 160000
B) 140000
C) 120000
D) 180000

Q33. Which salesperson (employee) handled the most orders?

```
SELECT e.emp_name, COUNT(o.order_id) AS total_orders
FROM employees e
JOIN orders o ON e.emp_id = o.emp_id
GROUP BY e.emp_name
ORDER BY total_orders DESC;
```

A) Ankit
B) Sneha
C) Rahul
D) Priya

Q34. Order 1004 was handled by a salesperson. Which department does that salesperson belong to?

```
SELECT d.dept_name
FROM orders o
JOIN employees e ON o.emp_id = e.emp_id
JOIN departments d ON e.dept_id = d.dept_id
```

```
WHERE o.order_id = 1004;
```

A) IT
B) Sales
C) HR
D) Finance

Q35. Which customers bought a Desk?

```
SELECT c.customer_name
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
JOIN products p ON o.product_id = p.product_id
WHERE p.product_name = 'Desk';
```

A) Ravi and Anjali
B) Anjali only
C) Ravi and Meena
D) Suresh and Anjali

Q36. How many orders were placed by customers from Delhi?

```
SELECT COUNT(o.order_id)
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
WHERE c.city = 'Delhi';
```

A) 2
B) 4
C) 3
D) 5

Q37. How many rows does this query return? (Note: the filter is inside ON.)

```
SELECT c.customer_name, o.order_id
FROM customers c
LEFT JOIN orders o
  ON c.customer_id = o.customer_id AND o.quantity > 1;
```

A) 3
B) 4
C) 6
D) 8

Q38. Now the filter is moved to WHERE. How many rows does this query return?

```
SELECT c.customer_name, o.order_id
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
WHERE o.quantity > 1;
```

A) 7
B) 4
C) 6
D) 3

Q39. Which department has the highest average salary?

```
SELECT d.dept_name, AVG(e.salary) AS avg_salary
FROM departments d
JOIN employees e ON d.dept_id = e.dept_id
GROUP BY d.dept_name
ORDER BY avg_salary DESC;
```

A) Finance
B) Sales
C) IT
D) HR

Q40. For customer Karan, what do COUNT(*) and COUNT(o.order_id) return in this query?

```
SELECT c.customer_name, COUNT(*), COUNT(o.order_id)
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_name;
```

A) 1 and 0
B) 0 and 0
C) 1 and 1
D) 0 and 1